DAYANANDA SAGAR UNIVERSITY

SCHOOL OF ENGINEERING

USN No: [illegible]

II Semester B. Tech. Mid Semester Examination
Computer Science and Engineering

Course Title: C Programming for Problem Solving (CPPS)
Duration: 75 Mins.
Time: 1.00 to 2.15 PM

Course Code: 25EN1114
Date: 14-03-2026
Max Marks: 40

Instructions to Candidates:

- Answer all the four Questions
- Each question carries 10 marks

		Marks	BT Level	CO's	PO's
Q. 1(a)	ii. Which loop is guaranteed to execute at least once? B) for loop B) while loop C) do-while loop D) None iv. Which operator is to find the remainder of division? A) / B) % C) * D) & iii. What is the value of x in this expression X=10 + 5 * 2? A) 30 B) 20 C) 25 D) 15 v. Which function is used to compare two strings in C? A) cmpstr() B) compare() C) strcomp() D) strcmp() i. Which of the following is a valid identifier in C? A) my_variable1 B) 2ndVariable C) float D) name!	05	2	1, 2	1

Q. 1(b)	**Predict the output of the following code snippet?**
	#include<stdio.h>
	int main() {
	int i;
	for (i=1; i<=5; i++) {
	if(i == 3) {
	break;
	}
	printf("%d ", i);
	}
	return 0; }
	2 4 5
	1, 2, 3
	
	#include<stdio.h>
	int main() {
	int x = 4;
	switch(x) {
	case 1: printf("One");
	break;
	case 2: printf("Two");
	break;
	default: printf("Other");
	}
	return 0; }
	other
		05	3	1	2

Q. 2(a)	**Construct** a C program to generate the roots of a quadratic equation. The various parameters for the quadratic equation are to be received as inputs from the user.	05	2	1	3
Q. 2(b)	**Evaluate** the following arithmetic and conditional C expressions, when a=2, b=4, c=3: iii. Z = c * a % b + 1 iv. X = (a + b) > c && a < b	05	3	1	2
Q. 3(a)	**Develop** a C program for the scenario, where A teacher wants to analyse the performance of students in a recent exam. The scores of 10 students are stored in a 1D array. The teacher wants to the Count of how many students scored above 75. Also Find the average score.	05	4	2	3
Q. 3(b)	**Predict** the output of the following code snippet?	05	3	2	2

```c
#include <stdio.h>
int main() {
    int arr[] = {3, 6, 9, 12};
    for (int i = 3; i >= 0; i--)
        printf("%d ", arr[i]);
    return 0;
}
```

12, 9, 6, 3

```c
#include <stdio.h>
#include <string.h>
int main() {
    char str[] = "Data";
    printf("%d ", sizeof(str));
    printf("%d", strlen(str));
    return 0;
}
```

5 4

Q. 4(a)	Illustrate the following string handling built-in functions and display the output. i. strcpy() ii. strcat() iii. strlen() iv. strcmp() v. strchr()	05	3	2	3
Q. 4(b)	Develop a C program to implement the addition of two matrices.	05	4	2	3

40+5
200

40+40+36+35

160+40
200

SCHOOL OF ENGINEERING

USN: ENG25CS0453

II Semester B.Tech Mid Semester Examination

Department of Mathematics

Course Title	:	Single and Multivariate Calculus	**Course Code**	:	25EN1201
Duration	:	75 Mins.	**Date**	:	13/03/2026
Time	:	9.00 AM - 10.15 AM	**Max. Marks**	:	40

Instructions to Candidates: 1. Answer all the questions. 2. Each full question carries 10 marks.

Q.No.		Question Description	Marks	BT Level	COs
1.	a)	Using chain rule, express $\frac{dw}{dt}$ in terms of t, where $w(x,y,z) = x^2 + xy + z^2,\ x = 2t,\ y = t^2,\ z = \sin t$.	4	L2	CO1
1.	b)	Find the absolute maximum and minimum values of $f(x,y) = x^2 + y^2 - 4x - 2y + 1$, on the triangular region in the first quadrant bounded by $x = 0,\quad y = 0,\quad y = 4 - x$.	6	L3	CO1
2.	a)	Test the continuity of the function at the origin $f(x,y) = \begin{cases} \frac{x^2y}{x^4+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0). \end{cases}$	4	L2	CO1
2.	b)	Applying Lagrange's multiplier method, find the extreme values of the function $f(x,y) = 4xy$ subject to the constraint $\frac{x^2}{8} + \frac{y^2}{16} = 1$.	6	L3	CO1
3.	a)	Evaluate the double integral $\iint_R (3x^2 + 2y)\, dA$, where R is bounded by $0 \leq x \leq 2,\quad 0 \leq y \leq 2x$.	5	L4	CO2
3.	b)	Using polar coordinates, evaluate the integral $\iint_R (x^2 + y^2) dA$, where R is the semicircular region bounded by the x-axis and the curve $y = \sqrt{1 - x^2}$.	5	L4	CO2
4.	a)	Solve $\int_0^1 \int_0^{1-x} \int_0^{x-y} dz\, dy\, dx$.	5	L3	CO2
4.	b)	Evaluate the triple integral in the cylindrical coordinates $\int_0^3 \int_0^{2\pi} \int_0^{\cos\theta} r dr d\theta dz$.	5	L4	CO2

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